Polar Equations and Graphs

ANSWER KEY

Section 1: Circles and Lines in Polar Form

1 What is the graph of $r = a$ where a is a constant?

a. **A circle centered at the origin with radius $|a|$** ✓

b. A line through the origin

c. A parabola

d. An ellipse

$r = a \Rightarrow \sqrt{x^2 + y^2} = a \Rightarrow x^2 + y^2 = a^2$, a circle centered at $(0, 0)$.

2 What is the graph of $\theta = \alpha$ where α is a constant angle?

a. **A straight line passing through the origin with slope $\tan\alpha$** ✓

b. A circle

c. A ray only

d. A spiral

Allowing all positive and negative values of r along the angle α produces a complete straight line through the origin.

3 Describe the circle given by $r = -6\sin\theta$.

a. **Centered at $(0, -3)$ with radius 3** ✓

b. Centered at $(0, 3)$ with radius 3

c. Centered at $(0, 0)$ with radius 6

d. Centered at $(-3, 0)$ with radius 3

Multiply by r: $r^2 = -6r\sin\theta \Rightarrow x^2 + y^2 = -6y \Rightarrow x^2 + (y+3)^2 = 9$. Center is $(0, -3)$, radius 3.

Section 2: Limaçons and Cardiods

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Under what condition does the limaçon $r = a + b\cos\theta$ $(a, b > 0)$ form a cardioid (heart-shaped curve)?

a. $a/b = 1$ **(i.e.** $a = b$**)** ✓

b. $a/b < 1$

c. $1 < a/b < 2$

d. $a/b \geq 2$

When $a = b$, the curve passes through the origin at a cusp, creating a cardioid.

5 Which equation produces a limaçon with an inner loop?

a. $r = 2 + 3\cos\theta$ ✓

b. $r = 3 + 3\cos\theta$

c. $r = 4 + 2\cos\theta$

d. $r = 3 + 2\cos\theta$

An inner loop occurs when $a < b$ (ratio $a/b < 1$). For $r = 2 + 3\cos\theta$, $a/b = 2/3 < 1$.

6 What type of curve is $r = 5 + 2\sin\theta$?

a. **A convex (dimple-free) limaçon** ✓

b. A cardioid

c. A limaçon with an inner loop

d. A dimpled limaçon

Here $a/b = 5/2 = 2.5 \geq 2$. When $a/b \geq 2$, the limaçon has no dimple and is convex.

Section 3: Rose Curves

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 **petals** ✓

b. 6 petals

c. 4 petals

d. 12 petals

For $r = a\cos(n\theta)$, if n is odd, the curve has exactly n petals. Here $n = 3$, so it has 3 petals.

8 How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 8 **petals** ✓

b. 4 petals

c. 16 petals

d. 2 petals

If n is even, the curve has $2n$ petals. Here $n = 4$ (even), so it has $2(4) = 8$ petals.

9 What is the maximum length of each petal in $r = 5\cos(2\theta)$?

a. 5 ✓

b. 10

c. 2.5

d. 2

The maximum value of $|r| = |5\cos(2\theta)|$ is 5, which determines the tip length of each petal.

Section 4: Lemniscates and Spirals

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 What figure-eight-shaped curve is given by $r^2 = 9\cos(2\theta)$?

a. **A lemniscate ✓**

b. A cardioid

c. A rose curve

d. A spiral of Archimedes

Equations of the form $r^2 = a^2\cos(2\theta)$ or $r^2 = a^2\sin(2\theta)$ represent lemniscates (infinity-symbol / figure-eight curves).

11 What type of curve is represented by the polar equation $r = 2\theta$ for $\theta \geq 0$?

a. **Spiral of Archimedes ✓**

b. Logarithmic spiral

c. Lemniscate

d. Circle

Equations where radius is directly proportional to angle ($r = a\theta$) are Archimedean spirals.

12 Which symmetry test verifies that a polar graph is symmetric with respect to the polar axis (the horizontal axis)?

a. **Replacing (r, θ) with $(r, -\theta)$ yields an equivalent equation ✓**

b. Replacing (r, θ) with $(-r, \theta)$

c. Replacing θ with $\theta + \pi$

d. Replacing r with $1/r$

Symmetry across the polar axis corresponds to reflection across $\theta = 0$, so replacing θ with $-\theta$ must leave the equation unchanged.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647